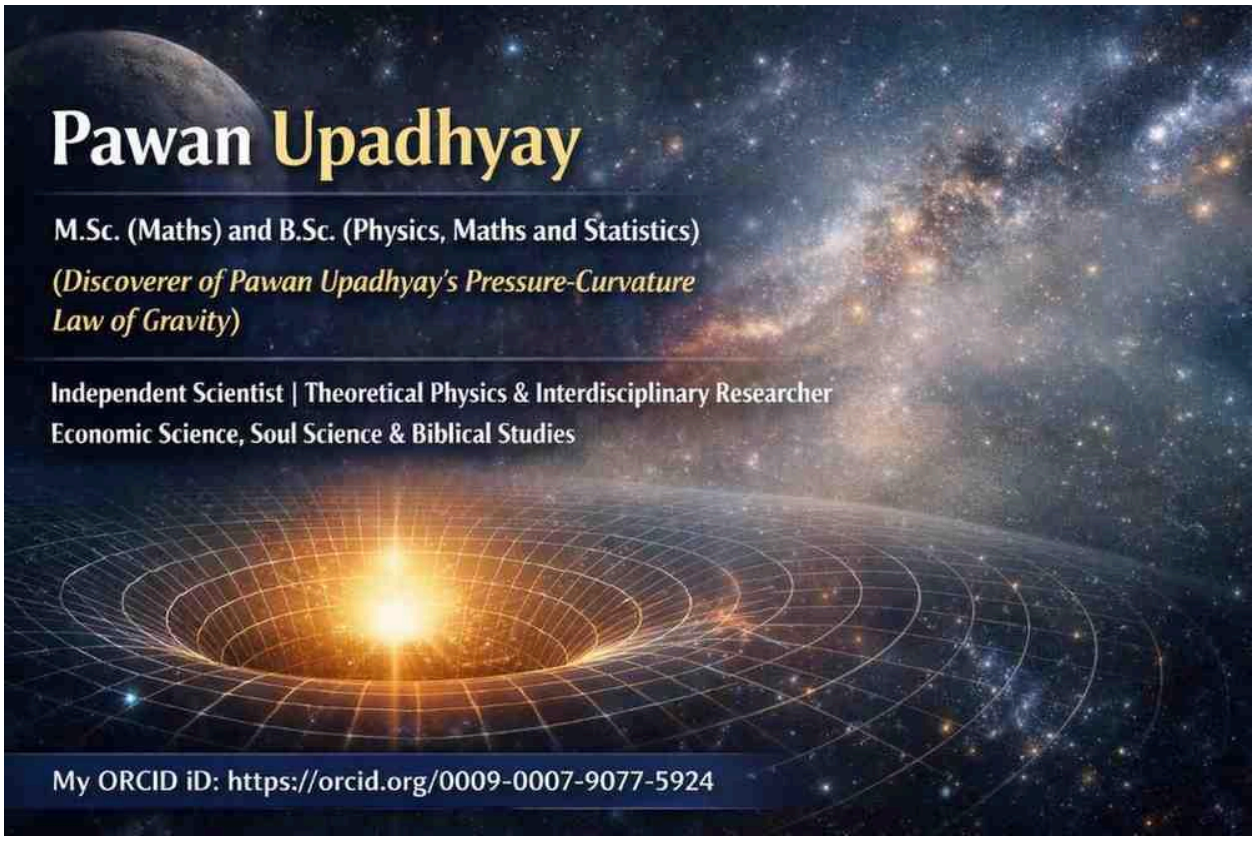
SYSTEM-DEPENDENT DEFINITION OF ENERGY DENSITY (E_d) AND MASS-EQUIVALENT DENSITY (ρ) IN THE PRESSURE–CURVATURE MODEL

A Theoretical Study of Ordinary Matter, Radiation, Non-Vacuum Energy, and Vacuum Energy

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ABSTRACT

The relationship between energy density and mass density is fundamental to the Pressure–Curvature (PPC) model. The proposed fundamental equation is:

$E_d = \rho c^2$

Here, E_d represents the energy density associated with the physical system under consideration, while ρ represents the corresponding mass density or mass-equivalent density.

This paper establishes that the definitions of E_d and ρ depend on the physical composition and scope of the system being investigated. The density may represent an individual material object, total local mass-equivalent density, total non-vacuum mass-equivalent density, or total mass-equivalent density including vacuum energy.

The paper examines the energy-density contributions associated with ordinary matter, radiation, dark matter, and vacuum energy. It distinguishes ordinary rest-mass density from total mass-equivalent density and explains why radiation energy and vacuum energy can contribute to mass-equivalent density even though they are not ordinary material rest mass.

The relationship is also examined in the context of the PPC equation:

$P_g = \omega E_d$

Particular attention is given to the vacuum-energy-like behaviour that occurs when $\omega = -1$.

Finally, the paper establishes the conditions under which the cosmological constant must be retained explicitly in the Friedmann equations and when its equivalent vacuum-energy density must instead be included in the total fluid energy density.

The principal conclusion is that the physical system determines the definition of ρ, and the corresponding energy-density definition determines the meaning of E_d.

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1. INTRODUCTION

Energy density and mass density are closely related through Einstein's mass-energy equivalence:

$E = mc^2$

For a physical system described in terms of densities, the corresponding relationship is:

$E_d = \rho c^2$

Where:

E_d = Energy density.

ρ = Corresponding mass density or mass-equivalent density.

c = Speed of light.

The Pressure–Curvature model uses this relationship as the foundation of its energy-density definition.

The proposed PPC equation is:

$P_g = \omega E_d$

Where:

P_g represents the pressure associated with the PPC relation, and ω represents its dimensionless equation-of-state parameter.

However, the physical meaning of E_d and ρ cannot be determined from the symbols alone. It depends on the physical system represented by those quantities.

For example, the density of an individual planet is different in physical meaning from the total mass-equivalent density of a cosmological region containing ordinary matter, radiation, and vacuum energy.

Similarly, ordinary rest-mass density is different from total energy density expressed as mass equivalent.

Therefore, a consistent PPC formulation must specify which physical contributions are included in E_d and which corresponding density is represented by ρ.

This paper develops that distinction systematically.

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2. FUNDAMENTAL MATHEMATICAL DEFINITION

The fundamental PPC energy-density relationship is:

$E_d = \rho c^2$

Rearranging:

$\rho = E_d/c^2$

This means that every energy-density definition has a corresponding mass-equivalent-density definition.

The SI units are:

Energy density:

$[E_d] = J/m^3$

Mass density:

$[\rho] = kg/m^3$

Speed of light:

$[c] = m/s$

Therefore:

$[\rho c^2] = (kg/m^3)(m^2/s^2)$

$[\rho c^2] = kg/(m \cdot s^2)$

$[\rho c^2] = J/m^3$

Thus, the equation is dimensionally consistent.

2.1. General interpretation

The equation does not require ρ to represent only ordinary matter rest-mass density.

Instead, ρ can represent the mass equivalent of whichever energy contributions are included in E_d.

For example:

If E_d represents ordinary matter rest-mass energy density, ρ represents the corresponding rest-mass density.

If E_d represents total non-vacuum energy density, ρ represents the corresponding total non-vacuum mass-equivalent density.

If E_d represents total energy density including vacuum energy, ρ represents the corresponding total mass-equivalent density including vacuum energy.

Therefore, the physical definition of ρ depends on the physical system represented by E_d.

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3. DISTINCTION BETWEEN ORDINARY MASS DENSITY AND MASS-EQUIVALENT DENSITY

An important distinction is necessary between ordinary rest-mass density and total mass-equivalent density.

3.1. Ordinary rest-mass density

Ordinary rest-mass density represents the rest mass of material constituents per unit volume.

It is commonly written as:

$$\rho_{rest} = m_{rest}/V$$

For ordinary nonrelativistic matter, the corresponding rest-mass energy density is approximately:

$$E_{rest} = \rho_{rest} c^2$$

However, an actual physical object may also contain:

1. Thermal energy.
2. Internal energy.
3. Kinetic energy.
4. Electromagnetic energy.
5. Elastic or nuclear binding energy.
6. Other contributions to its total energy.

Therefore, if ρ represents only ordinary rest-mass density, ρc^2 represents the rest-mass energy contribution, not automatically the complete local energy density of the object.

To represent the complete energy density, ρ must instead be defined as the corresponding total mass-equivalent density.

3.2. Total mass-equivalent density

Total mass-equivalent density is defined through the total energy density being considered:

$$\rho_{equiv} = E_{total}/c^2$$

Therefore:

$$E_{total} = \rho_{equiv} c^2$$

This definition can include rest-mass energy, radiation energy, internal energy, and other specified contributions.

In relativistic physics, the local energy density is associated with the time-time component of the stress-energy tensor in the chosen observer's rest frame. It is not generally identical to the conserved rest-mass density of material particles.

Consequently, the PPC model should identify ρ as ordinary rest-mass density or total mass-equivalent density whenever the equation $E_d = \rho c^2$ is applied.

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4. CASE I: ENERGY DENSITY OF AN INDIVIDUAL PHYSICAL OBJECT

Suppose ρ represents the mass density of an individual physical object.

Examples include:

1. Earth.
2. Sun.
3. Neutron stars.
4. Quark stars.
5. Other stars.
6. Other planets.

In this situation, the corresponding PPC energy density is:

$E_d = \rho c^2$

However, the precise interpretation depends on the definition of the object's density.

4.1. Earth

If ρ represents Earth's ordinary rest-mass density, then:

$E_{d,Earth} = \rho_{Earth} c^2$

This represents the rest-mass energy density associated with Earth's material density.

If ρ instead represents Earth's complete local mass-equivalent density, then E_d represents the corresponding total local energy density, including whichever internal and other energy contributions have been included in that definition.

4.2. Sun

For the Sun:

$E_{d,Sun} = \rho_{Sun} c^2$

The density must be specified as either ordinary rest-mass density or total mass-equivalent density.

The second definition can account for the additional energy contributions included in the selected physical model of the Sun.

4.3. Neutron stars

For a neutron star:

$E_{d,NS} = \rho_{NS} c^2$

The physical interpretation must distinguish rest-mass density from total local energy density.

This distinction is especially important in compact relativistic objects, where internal energy and pressure can make significant contributions to the stress-energy tensor.

4.4. Quark stars

For a hypothetical quark star:

$E_{d,QS} = \rho_{QS} c^2$

Again, the meaning depends on whether ρ represents the ordinary rest-mass density of the material or the complete mass-equivalent density associated with its energy content.

4.5. General conclusion for individual objects

For an individual physical object:

$E_{d,object} = \rho_{object} c^2$

The density definition determines whether this represents rest-mass energy density or total local energy density.

The PPC equation can then be written:

$P_{g,object} = \omega_{object} E_{d,object}$

The physical interpretation and value of ω must be established for the particular system.

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5. CASE II: TOTAL LOCAL MASS-EQUIVALENT DENSITY

Suppose ρ represents the total local mass-equivalent density of a physical system.

Then:

ρ = Total local mass-equivalent density.

The corresponding energy density is:

$E_d = \rho c^2$

Therefore:

E_d represents the total local energy density corresponding to that total mass-equivalent density.

This definition is broader than ordinary material rest-mass density.

The included energy contributions must be specified.

For example, a physical region may contain:

- Ordinary matter.
- Radiation.
- Dark matter.
- Internal energy.
- Other non-vacuum energy contributions.

If all these contributions are included, the total local energy density is the sum of the included energy densities.

The corresponding mass-equivalent density is obtained by dividing the total energy density by c^2.

Thus:

$\rho = E_d/c^2$

The term total local mass-equivalent density must not be confused with the ordinary rest-mass density of matter alone.

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6. CASE III: TOTAL NON-VACUUM MASS-EQUIVALENT DENSITY

Suppose ρ represents the total non-vacuum mass-equivalent density.

Then:

ρ = Total non-vacuum mass-equivalent density.

The corresponding energy density is:

$E_d = \rho c^2$

Therefore:

E_d represents the total non-vacuum local energy density.

The non-vacuum energy contributions may include:

1. Ordinary matter energy density.
2. Radiation energy density.
3. Dark matter energy density.
4. Other specified non-vacuum energy contributions.

In this definition, the vacuum-energy contribution associated with the cosmological constant is excluded.

For clarity, the energy-density budget can be represented as:

$E_{non\text{-}vacuum} = E_m + E_r + E_{dm} + E_{other}$

Where:

E_m = Ordinary matter energy density.

E_r = Radiation energy density.

E_{dm} = Dark matter energy density.

E_{other} = Other included non-vacuum energy contributions.

Therefore:

$E_d = E_{non\text{-}vacuum}$

And:

$\rho_{non\text{-}vacuum} = E_{non\text{-}vacuum}/c^2$

The corresponding PPC equation is:

$P_g = \omega E_d$

The parameter ω describes the pressure-to-energy-density relationship assigned to the PPC fluid under this definition.

It must not automatically be identified with the equation-of-state parameter of any one individual component.

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7. CASE IV: TOTAL MASS-EQUIVALENT DENSITY INCLUDING VACUUM ENERGY

Suppose ρ represents the total mass-equivalent density of a physical system that includes vacuum energy.

Then:

ρ = Total mass-equivalent density including vacuum energy.

The corresponding energy density is:

$E_d = \rho c^2$

In this case, E_d includes vacuum energy.

For example:

$E_d = E_m + E_r + E_{dm} + E_\Lambda + E_{other}$

Where:

E_Λ = Vacuum-energy density.

The corresponding mass-equivalent density is:

$\rho = E_d/c^2$

The vacuum-energy contribution has a corresponding mass-equivalent density:

$\rho_\Lambda = E_\Lambda/c^2$

This does not mean that vacuum energy is ordinary material rest mass.

It means that its energy density has a mass-equivalent representation through mass-energy equivalence.

Important mathematical consequence

If E_d includes vacuum energy, the same vacuum-energy contribution must not also be counted through an explicit cosmological-constant term in the geometric part of the Einstein field equations.

The two representations must remain consistent.

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8. CASE V: TOTAL ENERGY DENSITY INCLUDING ORDINARY MATTER, RADIATION, AND VACUUM ENERGY

Consider a system in which E_d represents the total local energy density of:

1. Ordinary matter.
2. Radiation.

3. Dark matter.
4. Vacuum energy.
5. Other specified energy contributions.

Then:

$E_d = E_m + E_r + E_{dm} + E_\Lambda + E_{other}$

The corresponding mass-equivalent density is:

$\rho = E_d/c^2$

Therefore:

$E_d = \rho c^2$

In this case, ρ represents the mass-equivalent density associated with the complete energy budget.

8.1. Ordinary matter contribution

The ordinary matter contribution can be represented as:

$E_m = \rho_m c^2$

Here, ρ_m is the corresponding rest-mass density in the nonrelativistic approximation.

8.2. Radiation contribution

Radiation has energy and momentum even though photons have zero invariant rest mass.

For an individual photon:

$E_\gamma = hv$

Where:

h = Planck constant.

v = Photon frequency.

For radiation in a physical volume V:

E_r = Total radiation energy/V

The corresponding mass-equivalent density is:

$\rho_r = E_r/c^2$

This is not photon rest-mass density. It is the mass-equivalent density of the radiation energy.

For isotropic radiation:

$P_r = E_r/3$

8.3. Vacuum-energy contribution

The vacuum-energy density associated with the cosmological constant is:

$E_\Lambda = \Lambda c^4/(8\pi G)$

Its corresponding mass-equivalent density is:

$\rho_\Lambda = E_\Lambda/c^2$

Therefore:

$\rho_\Lambda = \Lambda c^2/(8\pi G)$

Its pressure is:

$P_\Lambda = -E_\Lambda$

8.4. Combined energy density

If all these contributions are included:

$E_d = E_m + E_r + E_{dm} + E_\Lambda + E_{other}$

Then:

$\rho = (E_m + E_r + E_{dm} + E_\Lambda + E_{other})/c^2$

This provides a consistent definition of total mass-equivalent density for the selected energy budget.

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9. SYSTEM-DEPENDENT DEFINITIONS OF E_d AND ρ

The different cases can be summarized as follows.

Case 1: Individual physical object

ρ represents the density of an individual object, such as Earth, the Sun, a neutron star, or a quark star.

E_d represents the corresponding energy density.

If ρ is ordinary rest-mass density, E_d is rest-mass energy density. If ρ is total mass-equivalent density, E_d is total local energy density.

Case 2: Total local mass-equivalent density

ρ represents the total mass equivalent of the selected local energy budget.

E_d represents the corresponding total local energy density.

Case 3: Total non-vacuum mass-equivalent density

ρ represents the mass equivalent of all included non-vacuum energy contributions.

E_d represents total non-vacuum local energy density.

Case 4: Total mass-equivalent density including vacuum energy

ρ represents the mass equivalent of all included energy contributions, including vacuum energy.

E_d represents total local energy density including vacuum energy.

Case 5: Combined matter, radiation, and vacuum energy

ρ represents the mass equivalent of the combined energy budget.

E_d represents the total local energy density of the included ordinary matter, radiation, dark matter, vacuum energy, and other specified contributions.

The fundamental relationship remains:

$$E_d = \rho c^2$$

The physical interpretation changes according to the selected system.

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10. THE PRESSURE–CURVATURE EQUATION AND SYSTEM-DEPENDENT ENERGY DENSITY

The fundamental PPC equation is:

$$P_g = \omega E_d$$

Since:

$$E_d = \rho c^2$$

The equation can also be expressed as:

$P_g = \omega \rho c^2$

This relationship connects the pressure associated with the PPC model to the mass-equivalent density of the selected physical system.

The physical interpretation depends on:

1. The physical system being studied.
2. The definition of ρ.
3. The energy contributions included in E_d.
4. The meaning assigned to the PPC pressure P_g.
5. The equation-of-state parameter ω.

Therefore, the same mathematical relationship can be examined for different physical systems, but the physical interpretation of its variables must be specified in each application.

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11. EQUATION-OF-STATE PARAMETER FOR A COMBINED ENERGY SYSTEM

When E_d represents several physical energy contributions, the effective pressure-to-energy-density ratio must be defined carefully.

Suppose the total energy density is:

$E_{total} = E_1 + E_2 + E_3 + ... + E_n$

And the corresponding total pressure is:

$P_{total} = P_1 + P_2 + P_3 + ... + P_n$

Then the effective total equation-of-state parameter is:

$\omega_{total} = P_{total}/E_{total}$

Therefore:

$\omega_{total} = (P_1 + P_2 + ... + P_n)/(E_1 + E_2 + ... + E_n)$

This is generally not equal to the equation-of-state parameter of every individual component.

For example, in standard cosmology:

Pressureless nonrelativistic matter approximately has:

$\omega_m = 0$

Radiation has:

$\omega_r = 1/3$

Vacuum energy has:

$\omega_\Lambda = -1$

A combined system containing all three does not automatically have $\omega_{total} = -1$.

Its effective value depends on the relative energy densities and pressures.

For the PPC model, if ω describes the entire combined system, its physical definition must correspond to the pressure and energy density of that complete system.

If P_g is instead a separate PPC effective pressure, its relationship to the standard total pressure must be specified.

This distinction is essential for applying the PPC equation consistently.

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12. VACUUM-ENERGY-LIKE BEHAVIOUR WHEN ω = −1

The PPC equation is:

$P_g = \omega E_d$

For:

$\omega = -1$

We obtain:

$P_g = -E_d$

Therefore:

$P_g = -E_d$

This is the vacuum-energy-like equation-of-state relationship.

The standard cosmological-constant vacuum-energy relationship is:

$P_\Lambda = -E_\Lambda$

Both relationships have the same mathematical structure:

$P = -E$

Consequently, PPC produces vacuum-energy-like behaviour when its equation-of-state parameter is exactly −1.

However, this does not automatically establish:

$E_d = E_\Lambda$

The equality of the pressure-to-energy-density relationships does not prove that the energy densities have the same physical origin.

Two possible PPC definitions must be distinguished.

A. PPC energy density excludes vacuum energy

In this case:

E_d = Total non-vacuum local energy density.

The explicit cosmological constant remains a separate contribution.

If $\omega = -1$, the PPC fluid has an additional vacuum-energy-like equation of state.

B. PPC energy density includes vacuum energy

In this case:

E_d = Total local energy density including vacuum energy.

The PPC fluid's energy budget already includes E_Λ.

The corresponding cosmological equations must not count the same vacuum energy again through an explicit Λ term.

Thus, the meaning of $\omega = -1$ depends on the definition of E_d.

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13. COSMOLOGICAL CONSTANT AND THE TWO FRIEDMANN EQUATIONS

The treatment of the cosmological constant must be consistent with the selected definition of E_d.

13.1. When E_d excludes vacuum energy

The Einstein field equations retain an explicit cosmological constant:

$G_{\mu\nu} + \Lambda g_{\mu\nu} = (8\pi G/c^4)T_{\mu\nu}$

The first Friedmann equation is:

$H^2 = (8\pi G/3c^2)E_d - kc^2/a^2 + \Lambda c^2/3$

The second Friedmann equation is:

$\ddot{a}/a = -(4\pi G/3c^2)(E_d + 3P_g) + \Lambda c^2/3$

Using:

$P_g = \omega E_d$

We obtain:

$\ddot{a}/a = -(4\pi G/3c^2)E_d(1 + 3\omega) + \Lambda c^2/3$

Here, E_d excludes the vacuum-energy contribution represented by Λ.

No separate E_m and E_r terms are added if those contributions are already included in E_d.

13.2. When E_d includes vacuum energy

The cosmological constant can instead be represented as an effective fluid contribution.

In this formulation:

E_d (total) = E_d (non-vacuum) + E_Λ

P_g(total) = P_g (non-vacuum) + P_Λ

And:

$P_\Lambda = -E_\Lambda$

The Friedmann equations are then written without the corresponding explicit Λ term:

$H^2 = (8\pi G/3c^2)E_d(\text{total}) - kc^2/a^2$

$\ddot{a}/a = -(4\pi G/3c^2)(E_d \text{ total} + 3P_g \text{ total})$

These are two mathematically equivalent representations when the same cosmological constant is used consistently.

The explicit Λ representation and the vacuum-energy-fluid representation must not count the same contribution twice.

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14. CONSERVATION AND EVOLUTION OF ENERGY DENSITY

For a separately conserved homogeneous perfect fluid in an FLRW universe:

$\dot{E}_d + 3H(E_d + P_g) = 0$

Using:

$P_g = \omega E_d$

We obtain:

$\dot{E}_d + 3HE_d(1 + \omega) = 0$

For constant ω:

$E_d(a) = E_{d,0}\, a^{[-3(1 + \omega)]}$

This result applies to the selected conserved fluid.

Some important examples are:

A. Pressureless matter

$\omega = 0$

$E_d(a) = E_{d,0}\, a^{-3}$

B. Radiation

$\omega = 1/3$

$E_d(a) = E_{d,0}\, a^{-4}$

C. Vacuum-like PPC fluid

$\omega = -1$

$E_d(a) = E_{d,0}$

Therefore, for a separately conserved PPC fluid with $\omega = -1$, the energy density remains constant as the universe expands.

This is the same constant-density behaviour associated with cosmological-constant vacuum energy.

However, this conclusion requires separate conservation. If the PPC fluid exchanges energy with other components, an interaction term must be included.

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15. APPLICATION TO DIFFERENT PHYSICAL SYSTEMS

The system-dependent definition of E_d allows the PPC relation to be investigated in different physical settings.

15.1. Planetary systems

For a planet, the density may represent its ordinary material rest-mass density or a more complete local mass-equivalent density.

The corresponding energy density is:

$E_d = \rho c^2$

The choice must be stated.

15.2. Stellar systems

For ordinary stars, neutron stars, and other compact stellar objects, the density definition must account for the physical energy contributions included in the model.

In relativistic stellar structure, energy density and pressure are both relevant to the stress-energy tensor.

Therefore, ordinary rest-mass density alone is not necessarily sufficient to describe the complete gravitational source.

15.3. Radiation systems

For radiation, the corresponding mass-equivalent density is:

$\rho_r = E_r/c^2$

This does not assign invariant rest mass to photons.

It expresses the mass equivalent of the radiation energy.

15.4. Cosmological systems

For a cosmological region, E_d may represent the total non-vacuum energy density or the total energy density including vacuum energy.

The appropriate definition depends on whether the cosmological constant is retained explicitly or represented as part of the fluid.

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16. SCIENTIFIC LIMITATIONS AND REQUIREMENTS FOR FURTHER DEVELOPMENT

The equation:

$E_d = \rho c^2$

is a mass-energy relationship.

The equation:

$P_g = \omega E_d$

is an equation of state.

Together, they define a useful mathematical framework, but they do not by themselves establish a complete gravitational theory.

Further development requires:

1. A precise operational definition of ρ for each physical system.
2. A precise definition of which energy contributions are included in E_d.
3. A distinction between ordinary rest-mass density and total mass-equivalent density.
4. A definition of PPC pressure and its relationship to the physical pressure in the stress-energy tensor.
5. A complete conservation law for the PPC energy density.
6. A complete gravitational field equation or a clearly specified coupling to general relativity.
7. A derivation of the predicted behaviour of ω for different physical systems.
8. Quantitative comparisons with laboratory, astrophysical, and cosmological observations.

For compact objects, the distinction between local energy density, rest-mass density, pressure, and global gravitational mass must be handled using relativistic stellar structure.

For cosmology, the total equation-of-state parameter must be calculated consistently from the energy densities and pressures of the included components.

These requirements are necessary before the PPC model can be considered an independently established predictive gravitational theory.

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17. FINAL CONCLUSION

The fundamental relationship of the Pressure–Curvature model is:

$E_d = \rho c^2$

The corresponding PPC equation is:

$P_g = \omega E_d$

The definition of E_d and ρ depends on the physical system represented.

If ρ represents the ordinary rest-mass density of an individual object, E_d represents its corresponding rest-mass energy density.

If ρ represents the total local mass-equivalent density, E_d represents the corresponding total local energy density.

If ρ represents the total non-vacuum mass-equivalent density, E_d represents the corresponding total non-vacuum local energy density.

If ρ represents total mass-equivalent density including vacuum energy, E_d represents total local energy density including vacuum energy.

If E_d includes ordinary matter, radiation, dark matter, vacuum energy, and other specified energy contributions, ρ represents the mass-equivalent density corresponding to that complete energy budget.

Therefore, the same fundamental mathematical relationship can describe different physical systems, provided the physical meaning of its variables is explicitly defined.

When:

$\omega = -1$

The PPC equation becomes:

$P_g = -E_d$

This produces vacuum-energy-like behaviour. Nevertheless, the physical identity of E_d depends on whether vacuum energy is included in its definition.

FINAL DEFINING PRINCIPLE

The physical system determines what ρ represents, and the corresponding energy-density definition determines what E_d represents.

The general mathematical relationship remains:

$E_d = \rho c^2$

The Pressure–Curvature equation remains:

$P_g = \omega E_d$

The definition of E_d is system-dependent, while its relationship to the corresponding mass-equivalent density remains consistent.

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RESEARCH NOTE: SYSTEM-DEPENDENT ENERGY DENSITY, PPC TENSOR FORMULATION, AND GEODESIC MOTION IN THE PRESSURE–CURVATURE LAW OF GRAVITY

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ABSTRACT

The Pressure–Curvature (PPC) Law of Gravity employs an energy-momentum tensor formulation connecting energy density, gravitational pressure, and spacetime geometry.

Two fundamental relationships define the physical quantities used in the model:

$E_d = \rho c^2$

$P_g = \omega E_d$

The definition of E_d and ρ depends on the physical system under consideration. The density may represent the ordinary rest-mass density of an individual object, total local mass-equivalent density, total non-vacuum mass-equivalent density, or total mass-equivalent density including vacuum energy.

This note establishes an important mathematical distinction: changing the physical definition of E_d and ρ does not change the mathematical structure of the original PPC tensor equation.

The original PPC tensor is:

$T^{(PPC)}_{\mu\nu} = (E_d + P_g)u_\mu u_\nu + P_g g_{\mu\nu}$

Using the PPC pressure relationship and mass-energy equivalence, the tensor can be reformulated in terms of ρ and ω without changing its original tensor structure.

The note also incorporates the proposed PPC geodesic motion equation, in which the Christoffel symbols depend on the spacetime metric associated with E_d and P_g.

The principal conclusion is that the PPC tensor formulation is central to the mathematical structure of the Pressure–Curvature Law of Gravity. The definitions of E_d and ρ determine the physical interpretation of the tensor variables, while the tensor equation itself retains its mathematical form.

1. INTRODUCTION

The Pressure–Curvature Law of Gravity proposes a mathematical relationship between energy density and pressure:

$P_g = \omega E_d$

The energy density is related to its corresponding mass-equivalent density through:

$E_d = \rho c^2$

Where:

- E_d = Energy density represented by the selected physical system.

- ρ = Corresponding mass density or mass-equivalent density.
- P_g = Pressure associated with the PPC relationship.
- ω = Dimensionless equation-of-state parameter.
- c = Speed of light in vacuum.

The physical meaning of E_d and ρ depends on the physical system being investigated.

For example, the density of an individual planet may represent its ordinary rest-mass density, whereas the density of a cosmological region may represent the mass equivalent of ordinary matter, radiation, dark matter, and vacuum energy.

These different physical interpretations do not require different mathematical structures for the original PPC tensor.

The central objective of this note is to explain the relationship between:

1. System-dependent energy-density definitions.
2. The original PPC tensor equation.
3. The PPC pressure–energy-density relationship.
4. The reformulated PPC tensor.
5. The proposed PPC-dependent geodesic motion equation.

2. SYSTEM-DEPENDENT DEFINITION OF E_d AND ρ

The fundamental relationship is:

$E_d = \rho c^2$

Rearranging:

$\rho = E_d/c^2$

This expresses the mass-equivalent density corresponding to the selected energy density.

The SI units are:

$[E_d] = J/m^3$

$[\rho] = kg/m^3$

$[c] = m/s$

Therefore:

$[\rho c^2] = (kg/m^3)(m^2/s^2)$

$[\rho c^2] = J/m^3$

The equation is dimensionally consistent.

2.1. Ordinary rest-mass density

If ρ represents the ordinary rest-mass density of a physical object, then:

$\rho = \rho_{rest}$

And:

$E_d = \rho_{rest} c^2$

In this case, E_d represents the corresponding rest-mass energy density.

2.2. Total local mass-equivalent density

If ρ represents total local mass-equivalent density, then:

$\rho = E_{total}/c^2$

Therefore:

$E_d = E_{total}$

Here, the selected energy budget may include rest-mass energy, radiation, internal energy, and other specified energy contributions.

2.3. Total non-vacuum mass-equivalent density

If vacuum energy is excluded:

$E_d = E_{non\text{-}vacuum}$

The corresponding density is:

$\rho_{non\text{-}vacuum} = E_{non\text{-}vacuum}/c^2$

The included non-vacuum contributions may contain ordinary matter, radiation, dark matter, and other specified contributions.

2.4. Total mass-equivalent density including vacuum energy

If vacuum energy is included:

$E_d = E_m + E_r + E_{dm} + E_\Lambda + E_{other}$

The corresponding density is:

$\rho = E_d/c^2$

Here, E_Λ represents vacuum-energy density.

This is a mass-equivalent representation of energy. It does not mean that vacuum energy is ordinary material rest mass.

2.5. General principle

The physical system determines what ρ represents, and the corresponding energy-density definition determines what E_d represents.

The fundamental relationship remains:

$E_d = \rho c^2$

3. ORIGINAL PPC TENSOR EQUATION

The original PPC tensor equation is:

$T^{(PPC)}_{\mu\nu} = (E_d + P_g)u_\mu u_\nu + P_g g_{\mu\nu}$

Equation (13):

$T^{(PPC)}_{\mu\nu} = (E_d + P_g)u_\mu u_\nu + P_g g_{\mu\nu}$

The components are:

- $T^{(PPC)}_{\mu\nu}$ = PPC energy-momentum tensor.
- E_d = Energy density defined for the selected physical system.
- P_g = PPC pressure.
- u_μ = Four-velocity component.
- $g_{\mu\nu}$ = Spacetime metric tensor.

This is the original tensor formulation of the PPC model.

Its mathematical structure is established by the tensor equation itself.

4. PPC PRESSURE–ENERGY-DENSITY RELATIONSHIP

The PPC pressure relationship is:

$P_g = \omega E_d$

Substituting this relationship into the original tensor:

$T^{(PPC)}_{\mu\nu} = (E_d + \omega E_d)u_\mu u_\nu + \omega E_d g_{\mu\nu}$

Taking E_d as a common factor:

$T^{(PPC)}_{\mu\nu} = (1 + \omega)E_d u_\mu u_\nu + \omega E_d g_{\mu\nu}$

This is the first algebraic reformulation of the original PPC tensor.

5. REFORMULATION USING MASS-ENERGY EQUIVALENCE

The mass-energy equivalence relationship is:

$E_d = \rho c^2$

Substituting it into the reformulated PPC tensor:

$T^{(PPC)}_{\mu\nu} = (1 + \omega)\rho c^2 u_\mu u_\nu + \omega \rho c^2 g_{\mu\nu}$

Therefore:

$T^{(PPC)}_{\mu\nu} = (1 + \omega)\rho c^2 u_\mu u_\nu + \omega \rho c^2 g_{\mu\nu}$

This is the density-based representation of the original PPC tensor.

It expresses the tensor in terms of the selected mass-equivalent density and the PPC equation-of-state parameter.

6. IMPORTANT PRINCIPLE: DEFINITIONS OF E_d AND ρ DO NOT CHANGE THE TENSOR STRUCTURE

An important distinction in the PPC Law of Gravity is the difference between the physical definition of a tensor variable and the mathematical structure of the tensor equation.

The original tensor is:

$T^{(PPC)}_{\mu\nu} = (E_d + P_g)u_\mu u_\nu + P_g g_{\mu\nu}$

The fundamental definitions are:

$E_d = \rho c^2$

$P_g = \omega E_d$

These relationships determine the physical interpretation and mathematical relationships of the quantities appearing in the tensor.

They do not change the original tensor structure.

For example, E_d may represent:

1. Rest-mass energy density of an individual physical object.
2. Total local energy density.
3. Total non-vacuum energy density.
4. Total local energy density including vacuum energy.
5. Combined energy density of ordinary matter, radiation, dark matter, vacuum energy, and other specified contributions.

In every case, the original PPC tensor retains its form:

T^(PPC)_μv = (E_d + P_g)u_μu_v + P_g g_μv

The density-based reformulation also retains the same algebraic tensor structure:

T^(PPC)_μv = (1 + ω)ρc²u_μu_v + ωρc²g_μv

Thus, the system-dependent definition changes the physical interpretation of the variables, not the mathematical form of the original tensor.

7. CENTRAL IMPORTANCE OF THE TENSOR IN THE PPC LAW OF GRAVITY

In the Pressure–Curvature Law of Gravity, the tensor formulation matters most.

The PPC tensor provides the central mathematical structure through which the model expresses its energy-density and pressure relationships in tensor form.

The scalar relationships are:

E_d = ρc²

P_g = ωE_d

The tensor combines these quantities with the four-velocity and spacetime metric:

T^(PPC)_μv = (E_d + P_g)u_μu_v + P_g g_μv

The tensor therefore provides the mathematical framework in which the PPC physical variables are represented together.

The definitions of E_d and ρ establish what physical energy and density are represented in a particular application.

The pressure relationship establishes how P_g is related to E_d.

The tensor equation brings these quantities together in the PPC formulation.

Central PPC principle: The physical definitions of E_d and ρ may vary according to the selected system, but the original PPC tensor structure remains unchanged.

The tensor is the central mathematical object of the PPC formulation.

A complete gravitational theory must additionally specify the field equations and the relationship between the tensor and spacetime geometry.

8. PPC-DEPENDENT SPACETIME METRIC

The proposed PPC framework connects its energy-density and pressure variables with spacetime geometry.

This dependence is represented schematically as:

$g_{\mu\nu} = g_{\mu\nu}(E_d, P_g)$

Using:

$E_d = \rho c^2$

$P_g = \omega \rho c^2$

the metric dependence may also be represented as:

$g_{\mu\nu} = g_{\mu\nu}(\rho c^2, \omega \rho c^2)$

The Christoffel symbols are determined by the metric:

$\Gamma^{\mu}_{\alpha\beta} = (1/2)\, g^{\mu\sigma} (\partial_\alpha g_{\sigma\beta} + \partial_\beta g_{\sigma\alpha} - \partial_\sigma g_{\alpha\beta})$

PPC Geodesic Motion Equation: $d^2x^\mu/d\tau^2 + \Gamma^{\mu}_{\alpha\beta}(g_{\mu\nu}(E_d, P_g))\, (dx^\alpha/d\tau)(dx^\beta/d\tau) = 0$

Index definitions:

* μ = Free spacetime index.

* α and β = Summation indices.

* σ = Dummy summation index in the Christoffel-symbol equation.

* E_d = Energy density.

* P_g = PPC isotropic gravitational fluid pressure.

* g_μv = Spacetime metric.

* Γ^μ_αβ = Christoffel symbols.

* τ = Proper time.

These equations retain intended free index μ.

Consequently, if the PPC metric is determined by E_d and P_g, the Christoffel symbols inherit the dependence on these physical quantities.

This provides the proposed mathematical connection between PPC variables and spacetime geometry.

9. ORIGINAL PPC GEODESIC MOTION EQUATION

The proposed PPC geodesic motion equation is:

$d^2x^\mu/d\tau^2 + \Gamma^\mu_{\alpha\beta}(g_{\mu\nu}(E_d, P_g))(dx^\alpha/d\tau)(dx^\beta/d\tau) = 0$

This is the geodesic motion equation used in the PPC formulation.

Its components are:

- x^μ = Spacetime coordinate.
- τ = Proper time for a massive test particle.
- $\Gamma^\mu_{\alpha\beta}$ = Christoffel symbols.
- $g_{\mu\nu}(E_d, P_g)$ = PPC-dependent spacetime metric.
- E_d = Energy density of the selected physical system.
- P_g = PPC pressure.

The equation describes geodesic motion through the Christoffel symbols associated with the specified spacetime metric.

The PPC formulation proposes that the metric dependence is connected to its energy-density and pressure variables.

Using the PPC definitions:

$E_d = \rho c^2$

$P_g = \omega E_d$

the metric dependence can be expressed in terms of the selected mass-equivalent density and the equation-of-state parameter.

10. CONNECTION BETWEEN THE PPC TENSOR AND GEODESIC MOTION

The proposed PPC mathematical framework connects the following relationships.

10.1. Energy-density definition

$E_d = \rho c^2$

10.2. Pressure relationship

$P_g = \omega E_d$

10.3. Original PPC tensor

$T^{(PPC)}_{\mu\nu} = (E_d + P_g)u_\mu u_\nu + P_g g_{\mu\nu}$

10.4. Density-based PPC tensor

$T^{(PPC)}_{\mu\nu} = (1 + \omega)\rho c^2 u_\mu u_\nu + \omega \rho c^2 g_{\mu\nu}$

10.5. PPC-dependent metric

$g_{\mu\nu} = g_{\mu\nu}(E_d, P_g)$

10.6. PPC geodesic motion

$d^2x^\mu/d\tau^2 + \Gamma^\mu_{\alpha\beta}(g_{\mu\nu}(E_d, P_g))(dx^\alpha/d\tau)(dx^\beta/d\tau) = 0$

Together, these equations express the proposed connection between:

- Physical energy-density definitions.
- PPC pressure.
- PPC energy-momentum tensor.
- Spacetime metric.
- Christoffel symbols.
- Geodesic trajectories.

11. MATHEMATICAL REQUIREMENTS FOR THE TENSOR–GEODESIC CONNECTION

The geodesic equation describes freely falling test-particle motion in a specified spacetime metric.

To establish that the PPC tensor generates the metric used in the geodesic equation, the PPC gravitational field equations must specify the relationship between:

$T^{(PPC)}_{\mu\nu}$

and

$g_{\mu\nu}$

The conservation properties and matter-coupling assumptions must also be specified.

In a theory using the standard metric coupling of general relativity, covariant conservation of the total energy-momentum tensor is an important consistency condition:

$\nabla_\mu T^{\mu\nu} = 0$

For a perfect fluid, this conservation equation generally contains both energy evolution and momentum evolution. Nonzero pressure gradients can produce non-geodesic fluid motion, even though freely falling test particles follow metric geodesics.

The four-velocity normalization and unit convention must also be specified consistently when using the original PPC tensor equation.

These requirements are part of establishing the complete dynamical connection. They do not change the original PPC tensor equation.

12. VACUUM-ENERGY-LIKE CASE

The PPC equation is:

$P_g = \omega E_d$

For:

$\omega = -1$

we obtain:

$P_g = -E_d$

This gives the vacuum-energy-like pressure relationship.

The standard cosmological-constant vacuum-energy relationship is:

$P_\Lambda = -E_\Lambda$

Both have the same mathematical equation-of-state structure.

However, their physical interpretation depends on the definition of E_d.

If E_d excludes vacuum energy, the cosmological constant may remain a separate contribution.

If E_d includes vacuum energy, the same contribution must not be counted again through an explicit cosmological-constant term.

This distinction is necessary when applying the PPC framework to cosmology.

13. FINAL SCIENTIFIC CLARIFICATIONS

The following principles summarize the intended PPC formulation:

1. E_d is related to its corresponding mass-equivalent density through $E_d = \rho c^2$.
2. The physical meaning of ρ depends on the system under investigation.
3. E_d may represent rest-mass energy density or a broader selected energy density.
4. The PPC pressure relationship is $P_g = \omega E_d$.
5. The original PPC tensor is $T^{(PPC)}_{\mu\nu} = (E_d + P_g)u_\mu u_\nu + P_g g_{\mu\nu}$.
6. The definition of E_d and ρ does not change the mathematical structure of the original tensor.
7. The density-based reformulation is an algebraically equivalent representation of that tensor.
8. The tensor formulation is central to the mathematical construction of the PPC Law of Gravity.
9. The proposed geodesic equation uses Christoffel symbols associated with the PPC-dependent metric.
10. A complete dynamical derivation requires the PPC field equations and appropriate conservation and matter-coupling conditions.

14. CONCLUSION

The Pressure–Curvature Law of Gravity employs the fundamental relationships:

$E_d = \rho c^2$

$P_g = \omega E_d$

Its original tensor equation is:

$T^{(PPC)}_{\mu\nu} = (E_d + P_g)u_\mu u_\nu + P_g g_{\mu\nu}$

The equivalent density-based representation is:

$T^{(PPC)}_{\mu\nu} = (1 + \omega)\rho c^2 u_\mu u_\nu + \omega\rho c^2 g_{\mu\nu}$

The proposed PPC geodesic motion equation is:

$\frac{d^2x^\mu}{d\tau^2} + \Gamma^\mu_{\alpha\beta}(g_{\mu\nu}(E_d, P_g))\left(\frac{dx^\alpha}{d\tau}\right)\left(\frac{dx^\beta}{d\tau}\right) = 0$

The definition of E_d and ρ is system-dependent. It can represent ordinary rest-mass energy density, total local energy density, non-vacuum energy density, or total energy density including vacuum energy.

However, these different definitions do not alter the mathematical structure of the original PPC tensor.

The tensor formulation remains the central mathematical element of the PPC framework, while the system-dependent definitions establish the physical interpretation of its variables.

FINAL DEFINING PRINCIPLES

First principle — Energy-density equivalence:

$E_d = \rho c^2$

Second principle — PPC pressure relationship:

$P_g = \omega E_d$

Third principle — Original PPC tensor:

$T^{(PPC)}_{\mu\nu} = (E_d + P_g)u_\mu u_\nu + P_g g_{\mu\nu}$

Fourth principle — Tensor invariance under variable definition:

The definition of E_d and ρ does not change the mathematical structure of the PPC tensor.

Fifth principle — Geodesic motion:

The proposed PPC geodesic equation describes trajectories through Christoffel symbols associated with the PPC-dependent spacetime metric.

The central principle of the Pressure–Curvature Law of Gravity is that the PPC tensor formulation matters most. The physical definitions of its variables determine the system being represented, while the original tensor structure remains unchanged.

PPC EFFECTIVE STRESS–ENERGY TENSOR

$$T_{\mu\nu}^{\mathrm{PPC}} = \left(E_d + P_g\right) u_\mu u_\nu + P_g\, g_{\mu\nu}. \tag{35}$$

Where:

- E_d = effective energy density of the PPC field
- P_g = effective pressure (gravitational pressure) of the PPC field
- u_μ = 4-velocity of the fluid element $\left(u_\mu u^\mu = -1\right)$
- $g_{\mu\nu}$ = spacetime metric tensor

PPC EINSTEIN FIELD EQUATION

$$G_{\mu\nu} = \frac{8\pi G}{c^4}\, T_{\mu\nu}^{\mathrm{PPC}}. \tag{36}$$

Where:

- $G_{\mu\nu}$ = Einstein tensor (geometry of spacetime)
- $T_{\mu\nu}^{\mathrm{PPC}}$ = PPC effective stress–energy tensor
- G = Newton's gravitational constant
- c = Speed of light in vacuum

$$G_{\mu\nu} = \frac{8\pi G}{c^4} T_{\mu\nu}$$

Substituting PPC tensor:

$$G_{\mu\nu} = \frac{8\pi G}{c^4} \left[(E_d + P_g) u_\mu u_\nu + P_g g_{\mu\nu} \right]$$

Geodesic Motion Equation

$$\frac{d^2 x^\mu}{d\tau^2} + \Gamma^\mu_{\alpha\beta} \left(g_{\mu\nu}(E_d, P_g) \right) \frac{dx^\alpha}{d\tau} \frac{dx^\beta}{d\tau} = 0.$$

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5. The Pressure–Curvature (PPC) equations and formulations presented in this research note by Pawan Upadhyay.

SIMPLE RESEARCH NOTE: EARTH, SUN, STARS, AND NEUTRON STARS ARE PHYSICAL MATTER

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1. Introduction

Earth, the Sun, stars, neutron stars, planets, and other astronomical objects are physical matter systems.

They are not separate from matter. They are physical objects composed of material constituents and possess mass, energy, and physical properties.

Examples include:

1. Earth and other planets.
2. The Sun and other ordinary stars.
3. White dwarf stars.
4. Neutron stars.
5. Other compact stellar objects.
6. Hypothetical quark stars.

2. Matter and Mass Density

These physical objects contain matter and have mass.

Their mass density can be represented as:

$\rho = M/V$

Where:

- ρ = Mass density.
- M = Mass of the physical system.

- V = Volume of the physical system.

For an individual object, the corresponding rest-mass energy density is:

$E_d = \rho c^2$

This relationship applies when ρ represents the corresponding rest-mass density.

If ρ instead represents total mass-equivalent density, E_d represents the corresponding total energy density.

3. Examples of Physical Matter Systems

3.1. Earth

Earth is a physical planetary body composed of solid, liquid, and gaseous materials.

Its mass density can be represented by:

ρ_{Earth}

Its corresponding rest-mass energy density is:

$E_{d,Earth} = \rho_{Earth} c^2$

3.2. Sun

The Sun is a physical star composed primarily of ionized hydrogen and helium plasma.

Its density can be represented by:

ρ_{Sun}

Its corresponding rest-mass energy density is:

$E_{d,Sun} = \rho_{Sun} c^2$

3.3. Ordinary stars

Stars are physical matter systems containing extremely hot plasma.

Their corresponding rest-mass energy density can be represented as:

$E_{d,star} = \rho_{star} c^2$

3.4. Neutron stars

Neutron stars are extremely compact astronomical objects containing ultra-dense matter.

Their corresponding density and energy density can be represented as:

$E_{d,NS} = \rho_{NS} c^2$

For accurate relativistic calculations, rest-mass density must be distinguished from total local energy density.

3.5. Other planets and compact objects

Other planets, white dwarfs, and hypothetical quark stars are also physical systems that can be studied through their respective mass and energy densities.

4. Application to the Pressure–Curvature Model

The PPC model uses:

$E_d = \rho c^2$

And:

$P_g = \omega E_d$

Therefore:

$P_g = \omega \rho c^2$

These equations can be investigated for different physical matter systems, including Earth, the Sun, ordinary stars, neutron stars, and other astronomical objects.

The meaning of ρ must be specified according to the physical system and the density convention being used.

5. Conclusion

Earth, the Sun, stars, neutron stars, planets, and other astronomical objects are physical matter systems.

They possess mass, energy, and density and can therefore be investigated using mass-energy relationships.

The fundamental PPC relationship remains:

$E_d = \rho c^2$

And the PPC pressure relationship remains:

$P_g = \omega E_d$

The physical system determines the density being represented, while the mathematical relationships remain applicable when their variables are defined consistently.

Central principle: Planets, stars, neutron stars, and other astronomical objects are physical matter systems and must be included when discussing matter in gravitational physics.

CLARIFICATION NOTES ON THE DEFINITION OF ENERGY DENSITY (E_d) AND MASS DENSITY (ρ) IN THE PRESSURE–CURVATURE MODEL

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1. INTRODUCTION

The fundamental Pressure–Curvature (PPC) equation is:

$P_g = \omega E_d$

The energy density used in this equation is defined by:

$E_d = \rho c^2$

Where:

- E_d = Energy density represented by the physical system.
- ρ = Corresponding mass density or mass-equivalent density.
- c = Speed of light.
- P_g = Pressure associated with the PPC relation.
- ω = Dimensionless equation-of-state parameter.

The definition of E_d and ρ depends on the physical system and the type of density being represented.

The symbols E_d and ρ must therefore be interpreted according to the physical composition and scope of the system under consideration.

The purpose of these notes is to establish a consistent definition of energy density and mass density for different physical systems.

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2. GENERAL RELATIONSHIP BETWEEN ENERGY DENSITY AND MASS DENSITY

The fundamental relationship is:

$E_d = \rho c^2$

This relationship expresses the energy density associated with a corresponding mass density.

The dimensions are:

Energy density:

$[E_d] = J/m^3$

Mass density:

$[\rho] = kg/m^3$

Speed of light:

$[c] = m/s$

Therefore:

$[\rho c^2] = (kg/m^3)(m^2/s^2)$

$[\rho c^2] = J/m^3$

Thus, the equation is dimensionally consistent.

However, the physical meaning of ρ depends on which mass density is being represented.

It may represent:

1. Ordinary matter mass density.
2. Total local mass-equivalent density.
3. Total non-vacuum mass-equivalent density.

4. Total mass-equivalent density including vacuum energy.
5. Mass-equivalent density corresponding to the combined energy density of matter, radiation, and vacuum energy.

Each definition represents a different physical system or energy budget.

---

3. CASE I: ρ AS THE TOTAL MASS DENSITY OF AN INDIVIDUAL PHYSICAL OBJECT

Suppose ρ represents the total mass density of an individual physical object, such as:

- Earth.
- Sun.
- Neutron star.
- Quark star.
- Other stars.
- Other planets.

In this situation:

ρ = Total mass density of the individual physical object.

The corresponding energy density is:

$E_d = \rho c^2$

Therefore:

E_d represents the total energy density associated with the mass density of that particular physical object.

For example:

A. Earth

If ρ represents the mass density of Earth, then:

$E_{d,Earth} = \rho_{Earth} c^2$

B. Sun

If ρ represents the mass density of the Sun, then:

$E_{d,Sun} = \rho_{Sun} c^2$

C. Neutron Star

If ρ represents the mass density of a neutron star, then:

$E_{d,NS} = \rho_{NS} c^2$

D. Quark Star

If ρ represents the mass density of a quark star, then:

$E_{d,QS} = \rho_{QS} c^2$

In each case, E_d represents the energy density associated with the corresponding object's mass density.

The physical system determines the value and interpretation of ρ and E_d.

Important clarification: In relativistic astrophysics, the appropriate density must be defined carefully. Rest-mass density, total local energy density, and gravitational mass density are not automatically identical. The relation $E_d = \rho c^2$ applies directly when ρ is defined as the corresponding mass-equivalent density.

---

4. CASE II: ρ AS THE TOTAL LOCAL MASS-EQUIVALENT DENSITY

Suppose ρ represents the total local mass-equivalent density of a physical system.

In that situation:

ρ = Total local mass-equivalent density.

The corresponding energy density is:

$E_d = \rho c^2$

Therefore:

E_d represents the total local energy density associated with that total local mass-equivalent density.

Here, ρ is not restricted to the ordinary rest-mass density of individual material objects.

It represents the mass equivalent of the total energy density included in the selected physical system.

The definition must specify which energy contributions are included.

---

5. CASE III: ρ AS THE TOTAL NON-VACUUM MASS-EQUIVALENT DENSITY

Suppose ρ represents the total non-vacuum mass-equivalent density.

In this situation:

ρ = Total non-vacuum mass-equivalent density.

The corresponding energy density is:

$E_d = \rho c^2$

Therefore:

E_d represents the total non-vacuum local energy density associated with that mass-equivalent density.

The physical system may include:

1. Ordinary matter energy.
2. Radiation energy.
3. Dark matter energy.
4. Other specified non-vacuum energy contributions.

The vacuum-energy contribution associated with the cosmological constant is excluded under this definition.

Thus:

E_d = Total non-vacuum local energy density.

And:

E_Λ = Vacuum-energy density associated with the cosmological constant.

These are separately defined contributions.

The combined energy density, when both are included in the same total energy budget, is:

$E_{total} = E_d + E_\Lambda$

The corresponding combined mass-equivalent density is:

$\rho_{total} = E_{total}/c^2$

---

6. CASE IV: ρ AS THE TOTAL MASS-EQUIVALENT DENSITY INCLUDING VACUUM ENERGY

Suppose ρ represents the total mass-equivalent density of a system that includes vacuum energy.

In that situation:

ρ = Total mass-equivalent density including vacuum energy.

The corresponding energy density is:

$E_d = \rho c^2$

Therefore:

E_d represents the total local energy density, including vacuum energy.

It may contain contributions from:

1. Ordinary matter.
2. Radiation.
3. Dark matter.
4. Vacuum energy.
5. Other specified energy contributions.

Under this definition, vacuum energy is included in E_d.

Consequently, if the cosmological constant is also retained explicitly in the geometric part of the Einstein field equations, its vacuum-energy contribution must not be counted a second time.

There are two consistent representations:

Representation A: Explicit cosmological constant

- Exclude E_Λ from E_d.
- Retain Λ explicitly in the Friedmann equations.

Representation B: Vacuum energy included in total fluid density

- Include E_Λ in E_d.
- Remove the corresponding explicit Λ term from the geometric side of the Friedmann equations.

These two representations must not double-count the same vacuum-energy contribution.

---

7. CASE V: E_d AS THE TOTAL LOCAL ENERGY DENSITY OF ORDINARY MATTER, RADIATION, AND VACUUM ENERGY

Suppose E_d represents the total local energy density containing:

1. Total ordinary matter energy density.
2. Total radiation energy density.
3. Total vacuum-energy density.
4. Other specified energy contributions.

Then:

$$E_d = E_m + E_r + E_\Lambda + E_{other}$$

Where:

E_m = Total ordinary matter energy density.

E_r = Total radiation energy density.

E_Λ = Total vacuum-energy density.

E_{other} = Other included energy contributions.

The corresponding mass-equivalent density is:

$$\rho = E_d/c^2$$

Therefore:

$$E_d = \rho c^2$$

In this case, ρ represents the total mass-equivalent density associated with the complete energy density E_d.

It is important to understand that photons have zero invariant rest mass but possess energy and momentum. Their radiation energy contributes to the total energy density and therefore to the corresponding mass-equivalent density.

Similarly, vacuum energy contributes to the mass-equivalent density when it is included in E_d.

Thus, ρ in this case is not simply the ordinary rest-mass density of matter. It is the mass-equivalent density corresponding to the complete energy budget.

---

8. RELATIONSHIP BETWEEN THE DIFFERENT DEFINITIONS

The following table summarizes the different interpretations.

Case I: Individual physical object

ρ = Mass density of Earth, Sun, neutron star, quark star, or another individual object.

E_d = Energy density associated with that object's mass density.

Case II: Total local mass-equivalent density

ρ = Total local mass-equivalent density.

E_d = Total local energy density associated with that density.

Case III: Total non-vacuum mass-equivalent density

ρ = Total non-vacuum mass-equivalent density.

E_d = Total non-vacuum local energy density.

Case IV: Total mass-equivalent density including vacuum energy

ρ = Total mass-equivalent density, including vacuum energy.

E_d = Total local energy density, including vacuum energy.

Case V: Combined ordinary matter, radiation, and vacuum energy

ρ = Mass-equivalent density associated with the combined energy budget.

E_d = Total local energy density of ordinary matter, radiation, vacuum energy, and other included contributions.

---

9. RELATIONSHIP WITH THE PRESSURE–CURVATURE EQUATION

The PPC equation is:

$P_g = \omega E_d$

Since:

$E_d = \rho c^2$

The PPC equation can also be expressed as:

$P_g = \omega \rho c^2$

This shows that the pressure associated with the PPC relation depends on:

1. The selected physical system.
2. The definition of its mass density ρ.
3. The corresponding energy density E_d.
4. The equation-of-state parameter ω.

Therefore, the PPC equation can be applied to different physical systems, provided the relevant density and pressure are defined consistently.

However, the physical interpretation of ω must be specified for each system.

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10. VACUUM-ENERGY-LIKE BEHAVIOUR WHEN ω = −1

The PPC equation is:

$P_g = \omega E_d$

For:

$\omega = -1$

We obtain:

$P_g = -E_d$

Therefore:

The PPC pressure becomes equal to the negative of its associated energy density.

This produces vacuum-energy-like behaviour.

The standard cosmological-constant vacuum-energy relationship is:

$P_\Lambda = -E_\Lambda$

Both have the same mathematical equation-of-state relationship.

However, the physical interpretation depends on the definition of E_d.

If E_d excludes vacuum energy, the PPC fluid produces a separate vacuum-like contribution.

If E_d includes vacuum energy, the PPC fluid's total energy density already contains that contribution.

Therefore, the definition of E_d must always be specified before interpreting the physical meaning of $\omega = -1$.

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11. APPLICATION TO THE FRIEDMANN EQUATIONS

For a homogeneous and isotropic universe, when E_d represents the selected total fluid energy density, the first Friedmann equation with an explicit cosmological constant is:

$$H^2 = (8\pi G/3c^2)E_d - kc^2/a^2 + \Lambda c^2/3$$

The second Friedmann equation is:

$$\ddot{a}/a = -(4\pi G/3c^2)(E_d + 3P_g) + \Lambda c^2/3$$

Using:

$$P_g = \omega E_d$$

We obtain:

$$\ddot{a}/a = -(4\pi G/3c^2)E_d(1 + 3\omega) + \Lambda c^2/3$$

Important condition:

These equations with explicit Λ require E_d to exclude the vacuum-energy contribution already represented by Λ.

If E_d includes that vacuum energy, the equivalent Λ contribution must instead be incorporated into the fluid and removed from the explicit geometric term.

The density convention must remain consistent throughout the derivation.

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12. IMPORTANT SCIENTIFIC CLARIFICATIONS

The following principles establish the intended meaning of the PPC variables:

1. E_d is defined through $E_d = \rho c^2$.

2. ρ does not have only one universal physical interpretation. Its definition depends on the physical system.

3. If ρ represents the mass density of an individual object, E_d represents the corresponding energy density of that object.

4. If ρ represents total local mass-equivalent density, E_d represents the corresponding total local energy density.

5. If ρ represents total non-vacuum mass-equivalent density, E_d represents total non-vacuum local energy density.

6. If ρ includes vacuum-energy mass equivalence, E_d also includes the corresponding vacuum-energy density.

7. If E_d includes ordinary matter, radiation, and vacuum energy, ρ represents the mass-equivalent density associated with all those contributions.

8. Vacuum energy must not be counted twice when the cosmological constant is retained explicitly.

9. The physical interpretation of ω depends on the physical system and the energy-density definition.

10. The equation P_g = ωE_d establishes an equation of state; a complete gravitational theory requires additional field equations and conservation laws.

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13. FINAL CONCLUSION

The fundamental PPC relationship is:

$E_d = \rho c^2$

And:

$P_g = \omega E_d$

The definition of E_d and ρ depends on the physical system they represent.

The mass density may represent an individual physical object, total local mass-equivalent density, total non-vacuum mass-equivalent density, or total mass-equivalent density including vacuum energy.

Correspondingly, E_d may represent the energy density of an individual object, total local energy density, total non-vacuum energy density, or total local energy density including vacuum energy.

Therefore, the PPC variables must always be interpreted according to the selected physical system and its specified energy contributions.

FINAL DEFINING PRINCIPLE

The physical system determines what ρ represents, and the corresponding energy-density definition determines what E_d represents.

The general relationship remains:

$$E_d = \rho c^2$$

The Pressure–Curvature equation remains:

$$P_g = \omega E_d$$

The definition of E_d is therefore system-dependent, while the mathematical relationship between energy density and its corresponding mass-equivalent density remains consistent.